NETWORK PACKET ANALYSIS

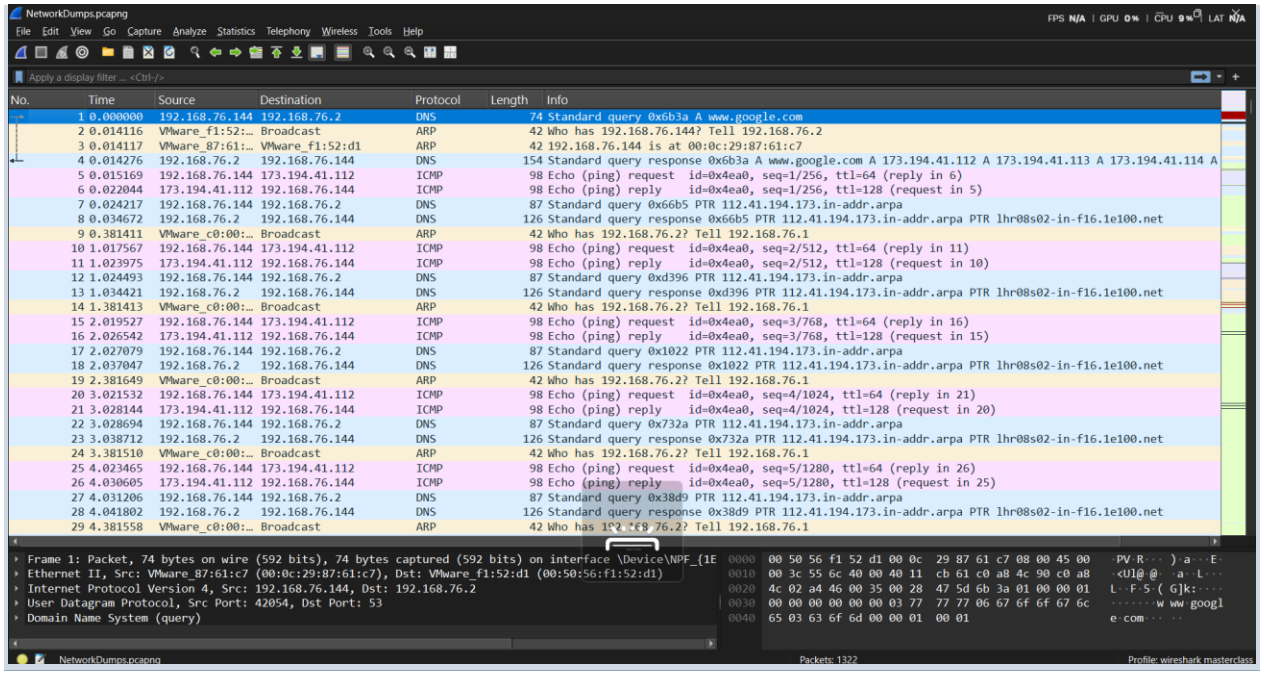
Lab Report & Forensics Walkthrough

File: NetworkDumps.pcapng | Tool: Wireshark

Field	Detail
Capture File	NetworkDumps.pcapng
Analysis Tool	Wireshark
Client IP	192.168.76.144
Gateway IP	192.168.76.2
Capture Date	10 February 2014
OS / Browser	Kali Linux 3.12 / Links 2.7 (text browser)

Section 1 — Overview & Environment

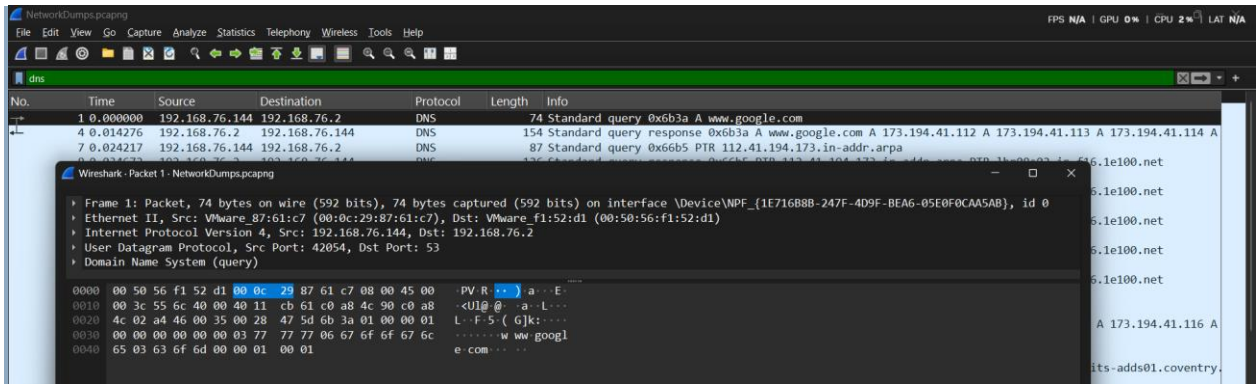
This lab involves analysing a real packet capture (PCAP) file to reconstruct the sequence of actions performed by a user on a networked machine. Using Wireshark, each packet was inspected to determine protocols, MAC addresses, HTTP interactions, and web-filtering behaviour.



Section 2 — Lab Questions & Answers

Q1. What was the first thing that the user did?

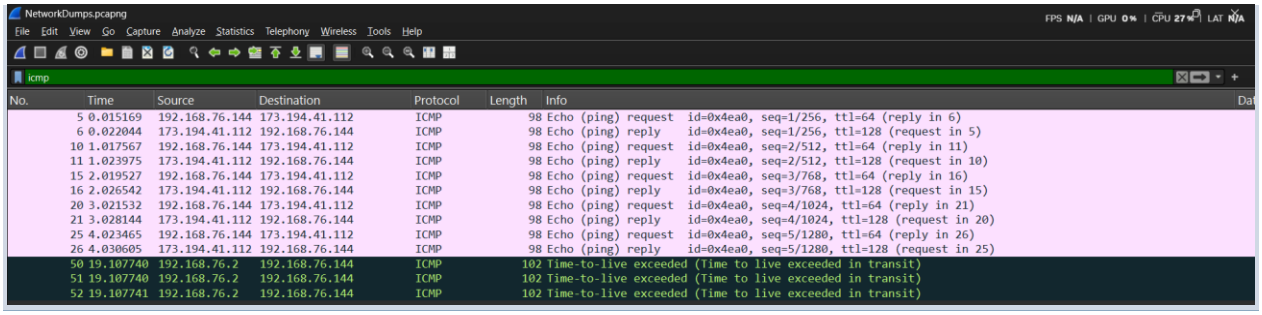
The user pinged www.google.com. Before the ping, the OS automatically issued a DNS query to resolve the hostname to an IP address (173.194.41.112). Five ICMP Echo Request packets were then sent to that IP.



Q2. What packet identifier(s) tell you this and why (i.e. name the protocol)?

Two protocols are involved:

Packets	Protocol	Purpose
1 & 4	DNS (Domain Name System)	Resolves www.google.com → 173.194.41.112 before the ping
5, 10, 15, 20, 25	ICMP (Internet Control Message Protocol)	Echo Request (type 8) — the actual ping packets sent to Google

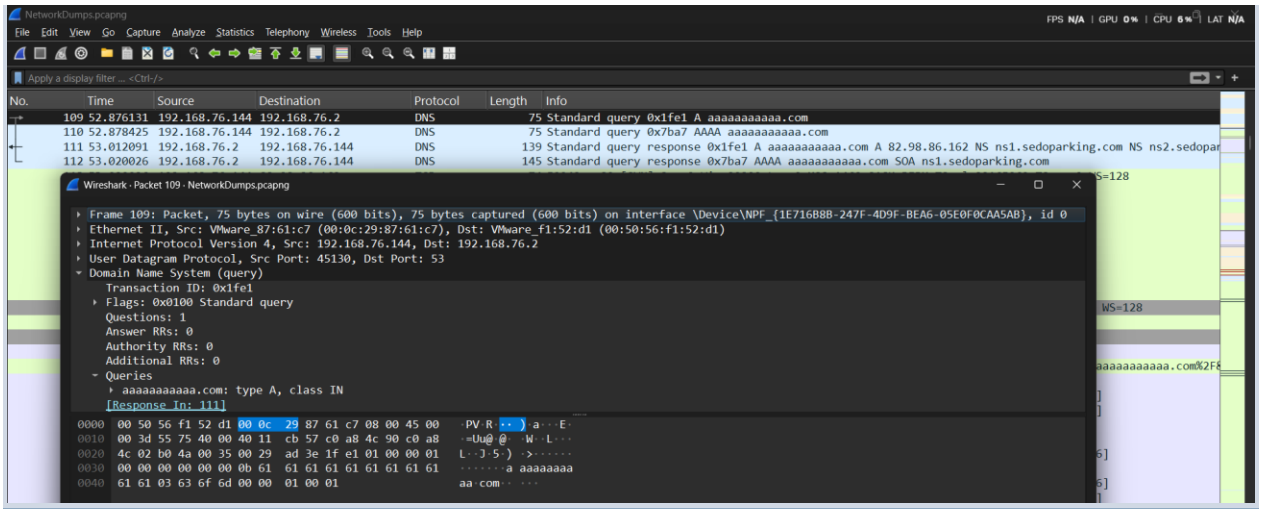


Q3. How many packets did the user send with regards to this specific protocol?

The user sent 5 ICMP Echo Request packets (packets 5, 10, 15, 20, and 25) to 173.194.41.112 (www.google.com). Each received a corresponding Echo Reply (type 0) from the server.

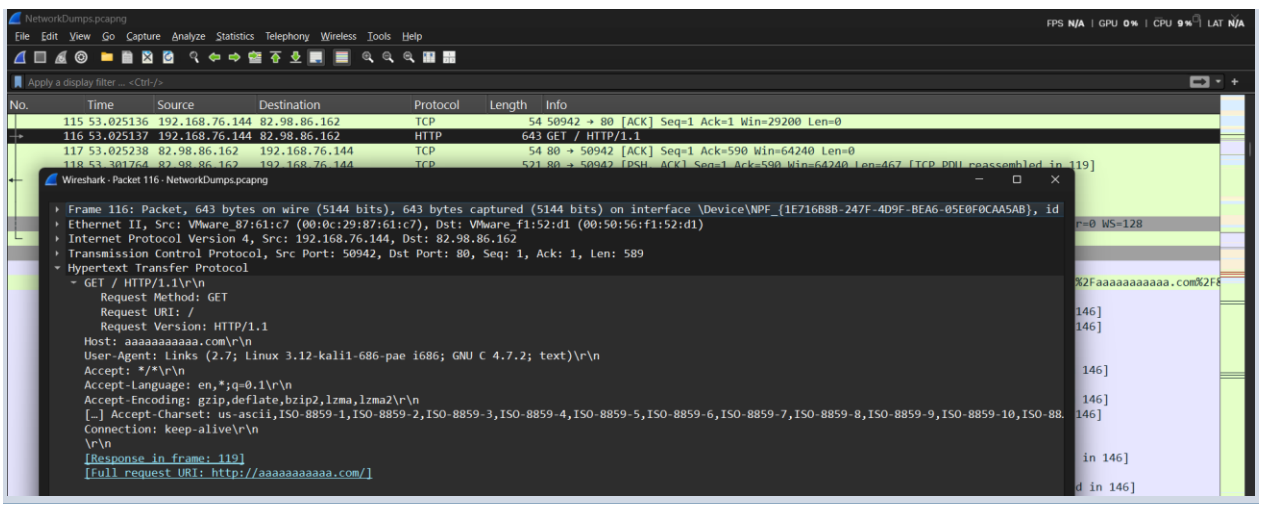
Q4. What was the second thing the user did and which website address did this involve?

The second activity was attempting to browse to http://aaaaaaaaaaa.com. DNS queries for this domain appear at packets 109–112, followed by an HTTP GET request at packet 116.



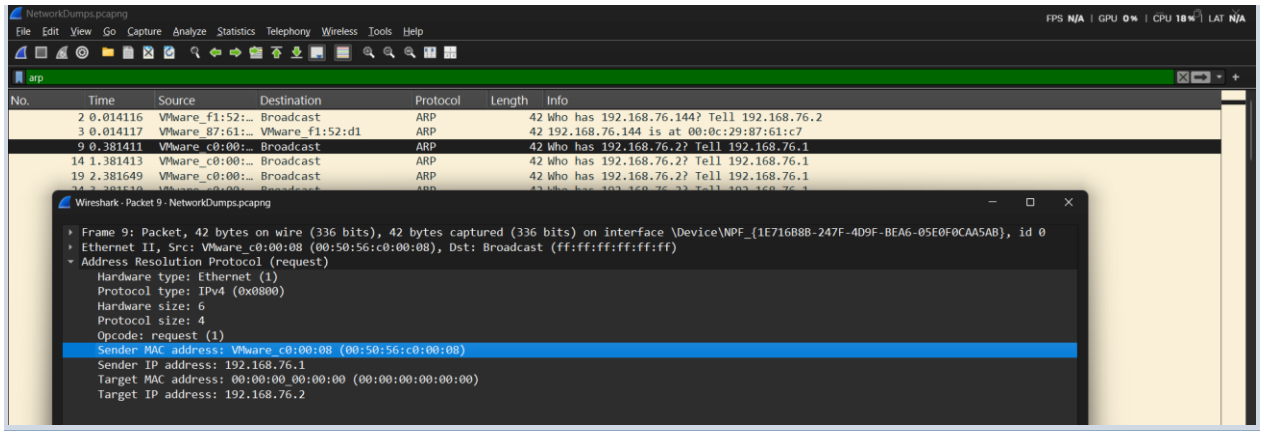
Q5. Which protocol would this be using?

HTTP (Hypertext Transfer Protocol) — specifically an unencrypted HTTP GET request sent over TCP port 80.



Q6. What is the MAC address of the machine with IP address 192.168.76.1?

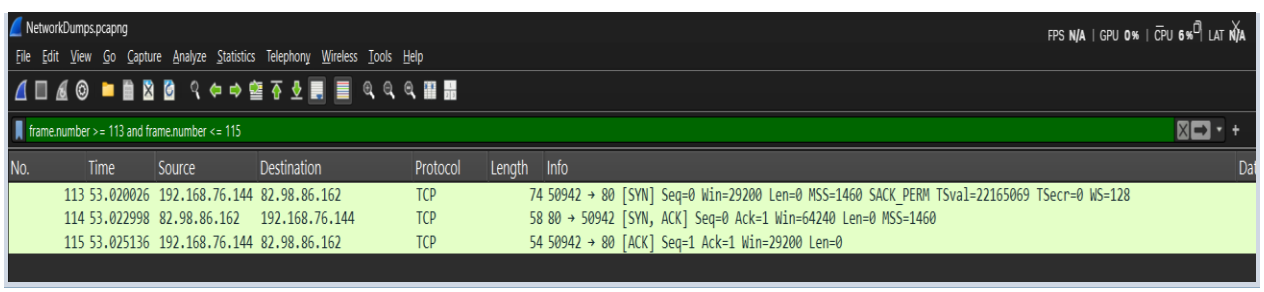
The MAC address is 00:50:56:c0:00:08. This was identified from ARP traffic. The OUI prefix 00:50:56 confirms this is a VMware virtual network adapter (acting as the default gateway).



Q7. Go to packet number 113. What connection-oriented interaction is taking place between packet number 113–115?

This is the TCP Three-Way Handshake — the process TCP uses to establish a reliable, connection-oriented session before any application data is exchanged.

Packet	Direction	TCP Flag	Meaning
113	Client (192.168.76.144) → Server (82.98.86.162)	SYN	"I want to connect" — client initiates
114	Server → Client	SYN, ACK	"Acknowledged, connect to me too" — server accepts
115	Client → Server	ACK	"Confirmed" — connection established



Q8. What website was the user trying to visit?

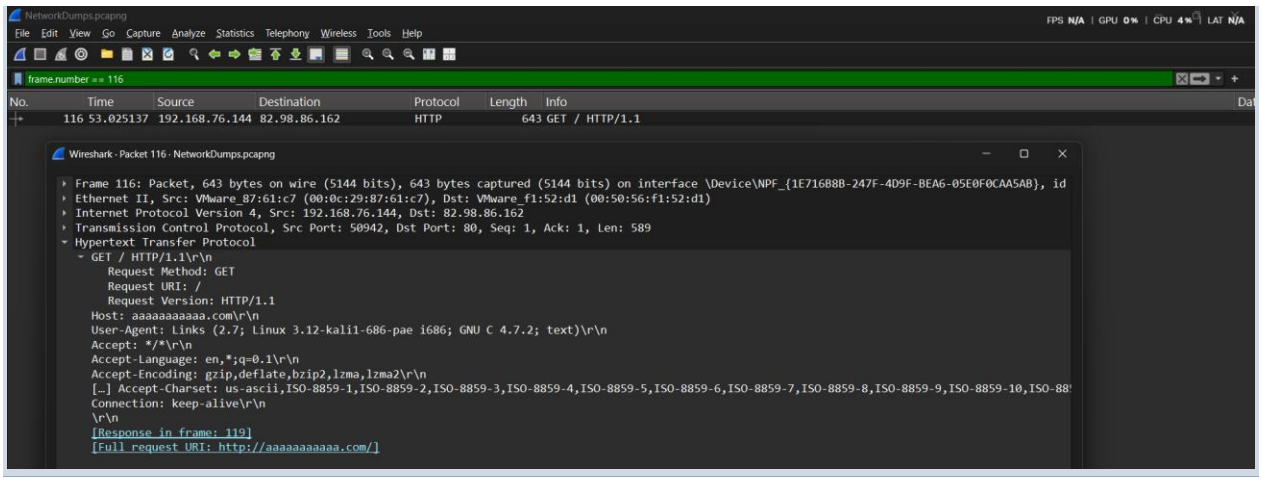
aaaaaaaaaaa.com — visible in the HTTP Host header of the GET request at packet 116, directed to IP 82.98.86.162 on TCP port 80.

Q9. What browser and OS is the user using to try and visit this website?

Extracted from the HTTP User-Agent header in packet 116:

```
User-Agent: Links (2.7; Linux 3.12-kali1-686-pae i686; GNU C 4.7.2; text)
```

Attribute	Value
Browser	Links 2.7 (a lightweight, text-based web browser)
OS / Kernel	Linux — Kali Linux, kernel 3.12-kali1-686-pae
Architecture	32-bit (i686)
Compiler	GNU C 4.7.2



Q10. Look at packet number 128. What does HTTP/1.1 200 mean?

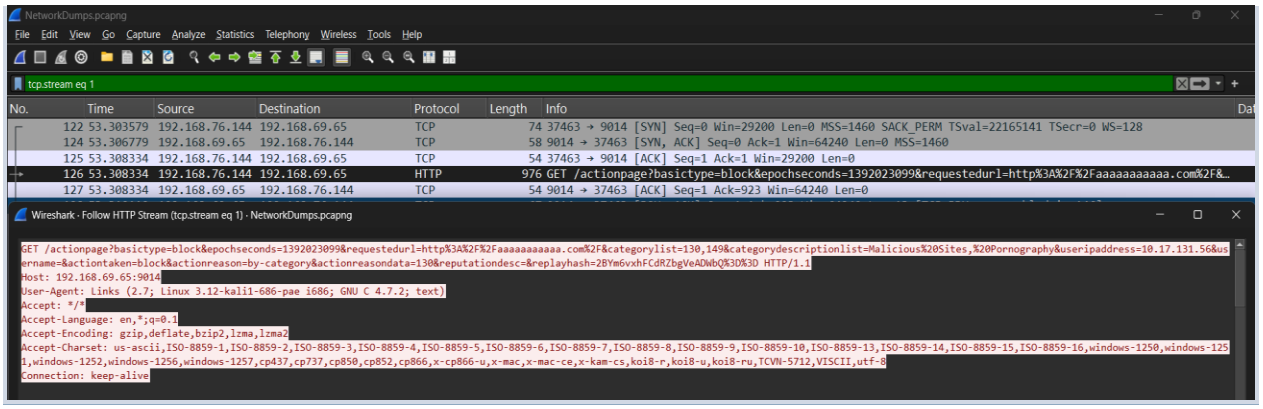
HTTP 200 OK is the standard HTTP success status code. It means the server successfully received the request and is returning the requested resource. In this context, packet 128 is part of the web content filter's response — the block page itself loaded successfully with a 200 OK, even though the original destination (aaaaaaaaaaa.com) was blocked.

Q11. Something unusual happened — instead of loading the page, it went to 'Cannot Access Webpage' (packet 138). Why might this be? (Hint: look at packet 126)

Packet 126 reveals a network-level web content filter intercepted the request. The browser was redirected to an internal filter server (192.168.69.65:9014) which logged the following block event:

```
requestedurl=http://aaaaaaaaaaa.com/
basictype=block
categorydescriptionlist=Malicious Sites, Pornography
categorylist=130,149
actiontaken=block
actionreason=by-category
```

The site aaaaaaaaaaa.com was categorised as both a Malicious Site (cat. 130) and Pornography (cat. 149). The filter blocked access and served the 'Cannot Access Webpage' denial page instead. This is consistent with enterprise or institutional web filtering products such as Websense / Forcepoint.

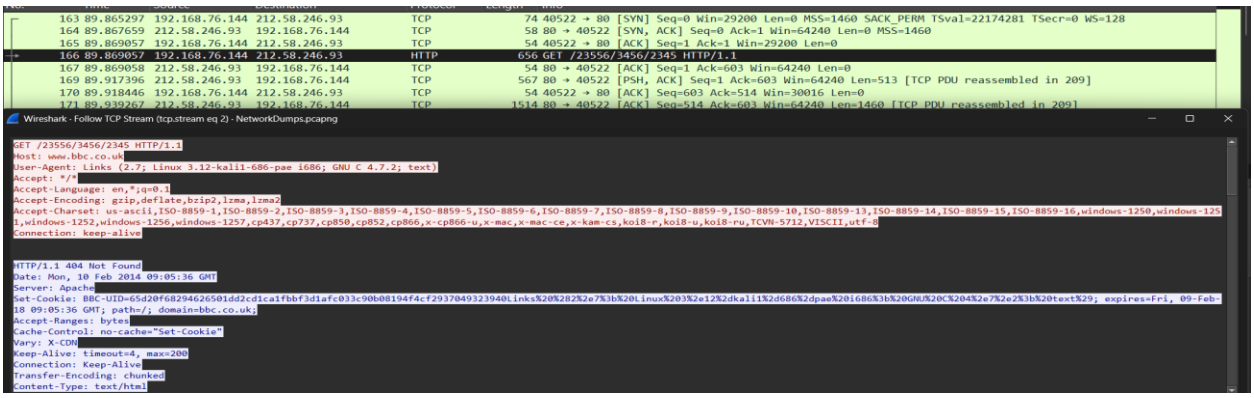


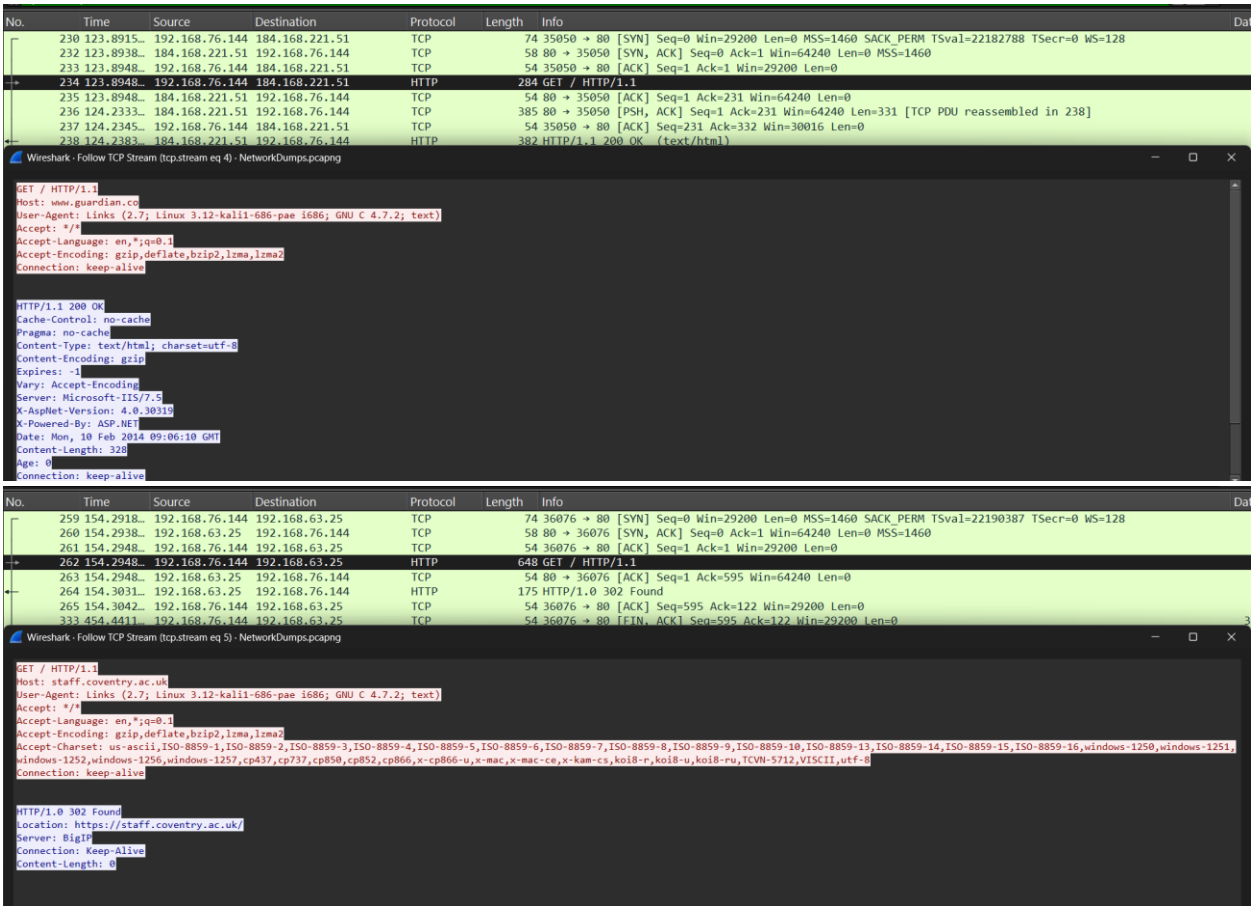
Q12. What was next in terms of website visited? (Give any details you can)

After the blocked site, three more websites were visited in sequence:

Order	Website	Packets	IP	Outcome
1st	www.bbc.co.uk	159–209	212.58.246.93	HTTP GET /23556/3456/2345 — received 404 Not Found (bad path)
2nd	www.guardian.co	219–239	184.168.221.51	DNS resolved via CNAME to guardian.co — HTTP 200 OK (page loaded)
3rd	staff.coventry.ac.uk	255–270	192.168.63.25	Internal server — HTTP 302 redirect → HTTPS port 443 → TLS Client Hello (encrypted session began)

The Coventry University staff portal (staff.coventry.ac.uk resolving to an internal 192.168.63.x IP) strongly suggests the user is on or affiliated with Coventry University. The site redirected HTTP to HTTPS, initiating a TLS handshake at packet 269.





Section 3 — Protocol Summary

Protocol	Role in This Capture	Key Packets
DNS	Resolves hostnames to IPs before each connection	1, 4, 7, 30–33, 109–112, 159–162, 219–222, 255–258
ICMP	Ping (Echo Request / Reply) to test connectivity	5, 6, 10, 11, 15, 16, 20, 21, 25, 26
ARP	Maps IP addresses to MAC addresses on LAN	2, 3, 9, 14, 19, 24, 29 …
TCP	Provides reliable, ordered transport (3-way handshake, teardown)	113–115, 223–225, 259–261 …
HTTP	Application-layer web requests (unencrypted)	116, 119, 126, 146, 166, 209, 226, 238, 262, 264
TLS	Encrypted HTTPS session (after 302 redirect)	269+

Wireshark · Protocol Hierarchy Statistics · NetworkDumps.pcapng

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	1322	100.0	1095708	17 k	0	0	0	1322
Ethernet	100.0	1322	1.7	18508	288	0	0	0	1322
Internet Protocol Version 6	1.1	15	0.1	600	9	0	0	0	15
User Datagram Protocol	1.1	15	0.0	120	1	0	0	0	15
Link-local Multicast Name Resolution	0.2	2	0.0	44	0	2	44	0	2
DHCPv6	1.0	13	0.1	1092	17	13	1092	17	13
Internet Protocol Version 4	92.0	1216	2.2	24320	379	0	0	0	1216
User Datagram Protocol	7.5	99	0.1	792	12	0	0	0	99
NetBIOS Name Service	0.5	6	0.0	354	5	6	354	5	6
Link-local Multicast Name Resolution	0.2	2	0.0	44	0	2	44	0	2
Dynamic Host Configuration Protocol	0.2	2	0.1	600	9	2	600	9	2
Domain Name System	2.9	38	0.2	2435	38	38	2435	38	38
Data	3.9	51	0.1	1632	25	51	1632	25	51
Transmission Control Protocol	83.5	1104	2.0	22272	347	1083	21852	341	1104
Transport Layer Security	0.6	8	0.9	9345	145	8	4769	74	9
Hypertext Transfer Protocol	1.0	13	77.6	849764	13 k	9	4256	66	13
Media Type	0.1	1	77.1	844288	13 k	1	844288	13 k	1
Line-based text data	0.2	3	3.0	33219	518	3	33219	518	3
Internet Control Message Protocol	1.0	13	0.1	844	13	13	844	13	13
Address Resolution Protocol	6.9	91	0.2	2548	39	91	2548	39	91

Section 4 — Key Findings & Learning Points

1. DNS precedes every connection — the OS resolves a hostname before any TCP handshake or HTTP request can occur.
2. ICMP (ping) is a diagnostic tool, not a browsing protocol — it uses type-8 Echo Requests and type-0 Echo Replies.
3. ARP operates at Layer 2 — it maps IP addresses to MAC addresses within the same LAN segment.
4. The TCP three-way handshake (SYN → SYN-ACK → ACK) must complete before any HTTP data is exchanged.
5. HTTP is plaintext — every request, response, User-Agent string, and Host header is readable in the capture.
6. Web content filters can intercept HTTP at the network level, silently redirecting blocked requests to a denial page — visible here via the /action page parameters in packet 126.
7. A 302 redirect signals that HTTP content has moved (temporarily) — often used to force an upgrade to HTTPS.
8. TLS (HTTPS) encrypts all subsequent traffic — the content of the staff.coventry.ac.uk session is not readable beyond the Client Hello.

This report was produced for educational and GitHub portfolio purposes. All analysis performed on a provided PCAP file using Wireshark.